

Hyperparameters and Seeds

Project: PPO-1 Autonomous Vehicle Parking

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1. Random Seeds

Purpose	Seeds
Training	42, 123, 456
Evaluation (held-out)	10000, 10001, 10002

All training seeds are held constant across hyperparameter settings. Hyperparameters are identical for every seed.

2. Full Hyperparameter Table (PPO – Stable-Baselines3)

These values are taken from `configs/ppo_parking.yaml` and were held constant across all three training seeds.

Hyperparameter	Value	Notes
Algorithm	PPO (Stable-Baselines3)	
Policy	MlpPolicy	
Network architecture (actor)	[256, 256]	
Network architecture (critic)	[256, 256]	
Activation function	ReLU	
Learning rate	3×10^{-4}	
n_steps	2048	
batch_size	64	
n_epochs	10	
Discount factor	0.99	Justified by short effective horizon of parking manoeuvres
GAE	0.95	
Clip range	0.2	
Entropy coefficient	0.01	Mild exploration
Value function coefficient	0.5	

Hyperparameter	Value	Notes
Max gradient norm	0.5	
Total timesteps per seed	300 000	May be reduced if compute is limited (state constraint in report)
Number of parallel environments	4	
Device	auto (CPU/GPU)	

Any hyperparameter not listed above uses the Stable-Baselines3 default.

3. Environment Configuration

Parameter	Value
Environment ID	parking-v0 (HighwayEnv)
Observation	Flattened KinematicsGoal (12-dimensional)
Action	ContinuousAction (steering + acceleration) $[-1, 1]^2$
Max episode steps	100
Reward weights	[1.0, 0.3, 0.0, 0.0, 0.02, 0.02]
Collision reward	-5.0
Success goal reward	0.12
Policy frequency	5 Hz
Simulation frequency	15 Hz

4. Evaluation Protocol

- Exploration disabled (`deterministic=True`)
- 30 evaluation episodes per seed
- Same evaluation seeds and metric code used for both the PPO agent and the rule-based baseline
- Mean and standard deviation reported across seeds for every metric

This document is part of the formal submission package required by DSCD 614.